

Governments as Design Contexts: Institutional Realities of Technology and Design in Government Settings

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Abstract

Governments at all levels use a dizzying variety of technology systems to coordinate work with each other, provide services, analyze data, and plan for the future. The use of these systems impacts all of us politically and personally, and their effective or ineffective functioning is bound up in immensely complex systems of power, ideology, and infrastructural investment. However, governments as design settings have been underexplored and undertheorized thus far in CSCW. In this one-day hybrid workshop, we will bring together a community of researchers interested in organizational and political analyses of technology design and implementation in the public sector. In so doing, we will examine current and emerging CSCW approaches to understanding organizational realities of tech development and implementation in the public sector and compile resources and a research agenda to facilitate future research.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing theory, concepts and paradigms; Collaborative and social computing design and evaluation methods.**

Keywords

government, technology, design, theory, method

ACM Reference Format:

Ridley Jones LeDoux, Seolha Lee, Pelle Tracey, Rachel B. Warren, Trine Rask Nielsen, and Andrew Hamann. 2025. Governments as Design Contexts: Institutional Realities of Technology and Design in Government Settings. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3715070.3748291>

1 Introduction and Background

Governments at all levels, from town councils to international federations, use a tremendous variety and volume of technology systems

to coordinate work with each other, provide services, analyze data, and plan for the future (e.g., [2, 44]). The outcomes of that work shape all human life and the broader terrestrial environment. This highly consequential, immensely complex genre of collaborative work has attracted attention in CSCW and HCI across various public sector domains, including child welfare [10, 36, 37], criminal justice [14, 39, 42], urban development [25, 29, 43], human services [3, 7, 18, 34], homeless services [30, 41], and asylum and migration [31, 32]. While many studies examine public data technologies and artifacts, governments as settings and government IT systems as design contexts have been underexplored and undertheorized in CSCW. The interaction forms, user types, jurisdictional issues, and institutional logics of technology development in government settings are distinctive in ways that create meaningful differences in theory and approach compared to other settings, an issue that has been identified from the field's early days [38]. As such, the relative lack of work in this area compared to that observing end user systems or corporate work settings is somewhat surprising. This workshop aims to bridge that gap by convening CSCW scholars working in government design and research settings.

Specifically, we call attention to the internal processes of technology development and implementation in government settings, as activities deeply embedded in the broader institutional fabric and history. Many aspects of these processes—procurement, contracting, and organizational relations and norms that govern such processes—are still relatively unknown to researchers and practitioners in CSCW. This is partly an access issue: for researchers outside governments, access to these processes is highly limited and often mysterious. CSCW researchers have engaged various government technology design projects, using the opportunities to critically examine the systems in power and transform them toward communities' benefit and social justice [18, 23, 37]. But often in these engagements, the scope of researchers' interactions has been confined to their purview as designers or facilitators, limiting their capacity to fully articulate the process and inform effective actions to influence them. As critical technical scholarship has argued, such disparity in research can also be sustained by researchers' own tendency to separate design, practice, and policy [19], or of failing to shift focus from discrete devices to "networks of working relations that make technical systems possible" [40]. Addressing these limitations is urgent under current conditions, where governments

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CSCW Companion '25, Bergen, Norway

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ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3748291>

struggle with a variety of new incursions and challenges to maintain existing services and infrastructure [35] as well as to begin new projects and ways of working in an increasingly fast-paced, privatized technology landscape [1, 47].

We will use this one-day workshop to examine current and emerging CSCW approaches to the organizational realities of public-sector technology development and implementation, compiling resources and an agenda to guide future research. We will situate contemporary government technologies in the longer history of public administration transformation driven by IT adoption and outsourcing [27, 47]. This will help us to grasp the broader landscape of government technology development and highlight understudied actors such as government IT professionals and private-sector contractors (Theme 1). We will also synthesize case studies from different political and organizational contexts to better understand the messiness of government tech development in practice (Theme 2). We seek inspirations from other related fields such as political science [12, 15], organization studies [13], and public administration [11] (Theme 3). Ultimately, our goal is to chart a CSCW research agenda to better understand and engage with the institutional realities of government technology development (Theme 4).

2 Workshop Themes

Below we describe a non-exhaustive set of key themes we expect to explore and develop during the workshop.

2.1 Situating government technologies in their historical contexts

Information technologies, from record-keeping and accounting to mapping and surveillance, have long been integral to government operation at all levels. Emergence of new IT tools and systems often corresponds to moments of flux, as government actors grapple with both potential affordances of new technologies and ideologies accompanying them. Adoption and development of such technologies have been intertwined with shifts in government structures and operations, whether through reconfigurations of organizational boundaries along the lines of expertise [5, 47] or new forms of social and legal control [26, 46]. These histories can help us better understand why technology and design seem to have the impacts they do on governments. For example, the rapid imaginative capture of “smart cities” as a site of “quantified community” [45] in recent years seems a new, even futuristic, intersection of design and government, raising new questions about governance and civic participation. Yet, aspects of its technocratic framing of civic life can be traced to earlier movements such as New Public Management [16, 27] and entrepreneurial governance [17].

In this theme, we examine lessons from earlier moments of change in government technology. We invite participants to consider:

- How have prior transformations of government structure and culture shaped the contemporary sociotechnical systems of government IT?
- What do histories of data governance have to tell us about the adoption of algorithmic tools in government?
- How have new forms of IT changed, or in some cases failed to change, the structures and cultures of government?

- What ideological positions about the roles of government are reified in particular instantiations of government technology?

2.2 Unpacking the messiness of technology development in institutions of governance

One of CSCW’s strengths as a field is its focus on the complex interplay between social and technical factors in technology development and deployment, reflected in extensive work on sociotechnical systems. As data-driven, algorithmic, and other IT systems have become increasingly salient in policy debate and public discussion, CSCW and related fields have examined government technologies as fundamentally sociotechnical [8, 24]. These approaches have expanded our empirical understanding of how government technologies work “in-practice” [14] and revealed unique design challenges posed by the institutional conditions of governments [20, 25, 28, 31].

With this second theme, we seek case studies that illuminate the organizational realities of technology development and implementation in government settings. These case studies will enable us to: 1) Compare case studies across varying domains, political contexts, and organizational cultures; 2) Learn new ways of formulating stakeholder needs and relationships; and 3) Better navigate the complex processes of introducing new data/information practice and system into the tightly woven, multiply committed network of governance.

Some questions participants may want to consider are:

- What unique forms and structures need to be designed with and for, such as jurisdiction? What design assumptions are currently bound up in US-centric models of government structure, legitimacy, and jurisdiction?
- CSCW researchers often participate as designers/technologists themselves in these research projects. What does this positionality implicate in doing critical research on pertinent organizational dynamics?
- What internal challenges do government actors experience in changing their data/tech practices? What can we learn about the more problematic aspects of technology transformations?

2.3 Theories and methods to rethink technology and design in governing organizations

In many cases, conventional design methods do not hold in the institutional settings of government without significant adaptation. Government decision-making processes are infrastructural—choices made “connect and constitute boundaries between public and private” [4]—and as a result, are highly path-dependent, constrained, and shaped by existing norms and relations prior to any design efforts. Sometimes “design” work can look much more like “implementation” or even “procurement” work, as a given form of a solution must be carefully adjusted to a rich, highly specific local context. Other fields such as public administration, organization studies, and political science may offer valuable approaches for making sense of these issues, and drawing on them can also provide CSCW, with its more deliberate focus on design, opportunities

to contribute to the work of these other fields. Some questions participants may want to consider are:

- How can we leverage theories and frameworks developed in the political science literature to better design and study collaborative technologies in government?
- How do theories from organization and public administration studies inform ways in which technologists can navigate and transform governing institutions and technical systems within those institutions?

2.4 Charting a CSCW research agenda for design in the public sector

Based on the discussions of the three preceding themes, we aim to chart a CSCW agenda for government technology development research and practice. Activities will include identifying research areas crucial but understudied in CSCW, as well as compiling resources from previous works in CSCW and other disciplines for future research. We also aim to rethink different models of technology and design in government settings to embrace their institutional realities. One such model is participatory design (PD) and European democratic design traditions. Rooted in the principle of involving stakeholders in design processes [6, 9, 21], PD methods have been critical tools for CSCW researchers who aim to responsibly design technologies within and for the public sector. Many CSCW scholars have experimented with PD methods for active engagement of frontline workers [18, 33] and other groups whose lives would be impacted by the adoption and transformation of technical systems [10, 22]. These studies have shown that participation in governments can operate differently with different stakeholders and settings, and such arrangements are often not given but a result of active work of negotiating access, building trust, and scaffolding stakeholders' agency beyond certain moments of design. Additional critical, participatory, and other traditions will guide and inspire, but not complete, our efforts. Questions participants may want to consider are:

- What other methods and strategies can be used to engage frontline workers and other stakeholders in CSCW research and design?
- How can we advance situated, context-sensitive, and democratic design practices in public sector CSCW research?

3 Workshop Goals and Outcomes

The primary goal of this workshop is to bring together a community of researchers with an interest in organizational and political analyses of technology design and implementation in the public sector. Connections built through this workshop will facilitate continued knowledge exchange among participants, and promote collaborations on projects among the participant researchers/practitioners and beyond.

We also aim to develop a CSCW research agenda to better understand and engage with the complexity of government technology ecosystems. The agenda will identify areas needing more scholarly and practical attention in CSCW, anticipate analytical and practical challenges, and highlight relevant methodological and theoretical approaches. Organizers will share it via the workshop website, blog posts, and potentially academic publications.

Lastly, we will compile interdisciplinary resources that can help technology researchers and practitioners better understand and engage with the complexity of government technology ecosystems. We will create a digital repository of resources collected during the workshop and share it publicly via a website.

4 Workshop Activities

4.1 Workshop attendees

We can accept up to 30 participants to ensure meaningful interactions. A call for participation will be shared via the workshop website, social media, mailing lists, and other channels.

Interested participants should provide a short bio (100 words) and statement of interest based on their past and future work. Statements may take one of three forms: (1) a write-up of up to 1,000 words (PDF), (2) a two-page visual presentation, or (3) a video of up to three minutes. Submissions should engage at least one workshop theme and include one conundrum or key question guiding their research related to the topic of this workshop. Questions under the workshop themes are examples of such conundrums. Organizers will use the conundrums to inform discussion items and breakout groups.

We seek submissions that 1) bring new theoretical perspectives on government technology; 2) suggest alternative imaginations and practices of government tech design that overcome the limitations of the existing ones; or 3) provide a rich case study of public-sector IT development, demonstrating the failures and successes of particular design approaches.

4.2 Workshop structure

This workshop will be fully hybrid. To address common challenges in hybrid workshops, we will survey participants about attendance plans. Remote attendees will be asked about technology and connectivity needs to configure hybrid and telepresence tools accordingly.

We will use a virtual whiteboard (e.g., Miro) to facilitate discussions between online and offline participants. A persistent group chat will be used before, during, and after the workshop to strengthen collaboration and connection between participants.

- Introduction (30 mins): Introductory presentation to set out the themes, agenda, and plans of the workshop.
- Lightning talks and short discussions (3.5 hours): Each participant will give a brief (up to 5 minutes) talk to recap their statement of interest. After every 5 to 6 talks, there will be a short synthesizing discussion. Participants will pose questions, comments, and feedback, which will be discussed further in the afternoon sessions.
- Lunch (1.5 hours)
- Breakout room discussions (2 hours): Participants will divide into breakout groups and discuss the inputs from the morning sessions across the four main themes.
- Plenary discussion (1 hour): All participants will reconvene to share what they discussed in breakout rooms, including key insights, and points of commonality and departure.
- Closing (30 mins): Summative discussion and initial plans for future development.

4.3 Equipment and supplies

To run this hybrid workshop, we will need a room with a hybrid conference setting, tables arranged for group discussion, a projector, a screen to display information from computers, whiteboards, dry markers, and post-it notes.

5 Organizers

Ridley Jones LeDoux is a PhD candidate in Human Centered Design & Engineering at the University of Washington. Her work focuses on interorganizational public service technology systems development, with a qualitative, ethnographically-informed approach.

Seolha Lee is a PhD candidate in Informatics at the University of California, Irvine (UCI). Her research combines ethnography and design research to investigate the roles and relations of actors across the public, private, and social sectors in shaping the government technology ecosystem and its political effects.

Pelle Tracey is a PhD candidate at the University of Michigan School of Information (UMSI). He is an ethnographer of record-keeping, broadly interested in how automated technologies, AI, and bureaucracy work in practice, particularly in frontline government or rental housing contexts.

Rachel B. Warren is a PhD student in Informatics at UCI. She uses a variety of methods to study predictive systems and data technologies in the public sector. She has worked with public defenders, data journalists, criminal justice reformers, and disaster planners. She is currently studying efforts to use data to optimize emergency medicine and emergency dispatching.

Trine Rask Nielsen is a postdoc in the Human-Centered Computing (HCC) section at the University of Copenhagen (UCPH) and a member of the Confronting Data Co-lab. Combining ethnography, participatory design, and research-through-design, she explores the role of data and data-driven technologies in public sector domains such as asylum, healthcare, and unemployment.

Andrew Hamann is a PhD student in Informatics at UCI. His research uses ethnography to study how techno-legal solutions frame public issue formation and possible futures. He is currently studying the downstream effects of data-intensive housing policy on IT infrastructure and labor in municipal government.

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